Do Now · retrieval
Hash tables with arrays

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a binary search tree, where are values smaller and larger than a node stored?

Q2

What does -1 represent in our pointer arrays?

Q3

What is a collision in hashing?

Q4

For table size 5, what bucket does key 17 map to using key MOD 5?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Create fixed buckets — CODE:
int[,] table = new int[5,3];
int[] count = new int[5]; — Each row is a bucket with three available collision slots.

Core idea 2

Hash — CODE:
int bucket = key % 5; — The same key maps to the same row.

Core idea 3

Add — CODE:
table[bucket, count[bucket]] = key;
count[bucket]++; — A collision goes into the next free column in that bucket.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Search/remove — CODE:
for (int c = 0; c < count[bucket]; c++)
{
if (table[bucket,c] == key)
{
// found
}
} — Only the calculated bucket is searched. Removal can shift later bucket items left.

Vocabulary · use the correct technical terms

hash table — A structure using a hash function to choose storage location.

key — The value supplied to the hash function.

hash function — A deterministic calculation producing an index.

bucket — A hash-table storage location.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Add 12, 17 and 22

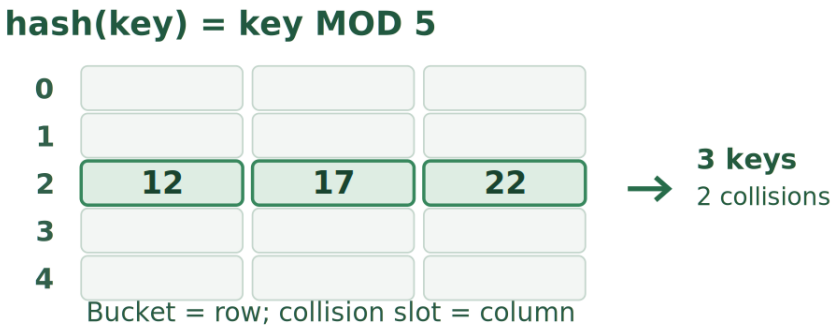
Follow the worked method

1. 12 — CODE:
table[2,0] = 12
count[2] = 1

2. 22 — CODE:
table[2,2] = 22
count[2] = 3

3. 17 — CODE:
hash → bucket 2
scan row 2
17 found at column 1

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same calculation
CODE:
37 MOD 10 = ?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Create buckets
Create a 5×3 integer table and a count array.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Remove 17
Find 17 in its bucket, shift later values left, and reduce the bucket count.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hash function tells you which bucket to search; it does not guarantee that only one key will use that bucket.

Use key vocabulary from memory

1. Explain hash table in your own words.

2. Explain key in your own words.

3. Explain hash function in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.